Persona Directions Reduce Forced-Choice Sycophancy via Activation Steering

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Abstract

Large language models often exhibit sycophancy, adapting responses to match a user's stated view even when it is false. We test whether general-purpose persona directions, extracted without sycophancy labels, reduce forced-choice sycophancy as effectively as a targeted Contrastive Activation Addition (CAA) baseline. On Gemma 2 27B and Qwen 3 32B, evaluated on a held-out PhilPapers A/B benchmark with Holm-corrected significance testing, critical-role vectors (Skeptic, Devil's Advocate, Judge) reach 68% (Gemma) and 116% (Qwen) of CAA's Δlogit and are Holm-significant on all test seeds; conformist roles show weak, heterogeneous effects. All role–CAA cosines are below 0.17, but the sign of $\cos(\text{role}, \text{CAA})$ flips across models, so geometric independence does not establish mechanistic independence. Critical-role steering is a lightweight, data-efficient intervention, while the polarity asymmetry complicates simple family-level predictions.

1. Introduction

Sycophancy — the tendency of large language models to agree with users regardless of factual correctness — is among the most persistent failure modes of RLHF-trained systems (Perez et al., 2022; Sharma et al., 2023). Even when a model has encoded the correct answer internally, the reward model's preference for agreement can override it (Wang et al., 2025a). How can we intervene on sycophancy without expensive retraining or curated behavioral datasets?

Contrastive Activation Addition (CAA) (Rimsky et al., 2024; Turner et al., 2023) extracts a steering vector from contrastive sycophantic/honest prompt pairs and adds a scaled version during inference. While effective, CAA requires hundreds of behavior-specific pairs and must be re-curated for each new target behavior. A natural alternative is to reuse persona representations already learned by instruction-tuned models (Lu et al., 2026).

Why roles matter. Beyond data-efficiency, interpretable role directions give deployers and auditors a vocabulary — "steer toward Judge" — that can be audited, composed with other personas, and contested, and a single persona library amortizes across many alignment-adjacent behaviors without behavior-specific contrast pairs for each.

We ask three questions. (i) Can off-the-shelf role vectors reduce forced-choice sycophancy comparably to CAA? (ii) Do critical/conformist family labels predict steering direction? (iii) Are effective role vectors geometrically distinct from the CAA direction? We evaluate across Gemma 2 27B (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) on a counterbalanced PhilPapers benchmark with tune/test splitting and Holm correction.

Contributions. (1) Critical-role vectors reach 68–98% of CAA's Δlogit with strong cross-seed consistency, using no sycophancy labels. (2) Conformist roles produce weak, heterogeneous effects, partially falsifying bidirectional family-level prediction. (3) All role vectors are nearly orthogonal to CAA ($|\cos| < 0.17$), but the sign of the cosine flips between Gemma and Qwen — a cross-model geometric asymmetry we make explicit as a new caveat on mechanistic-independence claims.

2. Related Work

Sycophancy in LLMs. Perez et al. (2022) introduced model-written sycophancy evaluations exposing systematic agreement bias. Sharma et al. (2023) showed sycophancy is widespread and emerges during RLHF.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

Wang et al. (2025a) traced sycophancy to override mechanisms in RLHF-trained models: the correct answer is often encoded internally but suppressed by preference for agreement.

Activation steering and CAA. Activation addition (Turner et al., 2023) demonstrated that fixed residual-stream vectors can steer LLM behavior at inference time. CAA (Rimsky et al., 2024) formalizes this via mean-difference contrasts on labelled A/B pairs. Related work includes representation engineering (Zou et al., 2023), inference-time intervention (Li et al., 2023), and conditional activation steering (Lee et al., 2025). Goral et al. (2025) study depth-wise steering across layers.

Persona directions. Lu et al. (2026) introduced the Assistant Axis and released per-role steering vectors (Skeptic, Judge, Peacekeeper, ...) derived from a generation-plus-judge contrastive pipeline. Feng et al. (2026) compose persona directions via vector algebra for inference-time personality control. Pai et al. (2025) merge persona vectors. Wang et al. (2025b) link persona features to emergent misalignment.

Persona–sycophancy link. Shah et al. (2026) showed persona agreeableness correlates with sycophancy at r up to 0.87. Vennemeyer et al. (2025) argue sycophancy decomposes into causally separable components along distinct linear directions. Our work sits at this intersection: we test whether off-the-shelf role vectors — never trained on sycophancy labels — transfer to a held-out forced-choice benchmark, and we characterize the geometric relationship to a targeted CAA direction on two models.

3. Methodology

3.1. Models and target layers

We use Gemma 2 27B Instruct (Gemma Team, 2024) and Qwen 3 32B (Qwen Team, 2025) because they are instruction-tuned decoder-only models at comparable scale but with substantially different baseline sycophancy rates on our benchmark (59% vs. 84%), providing a natural robustness test. Steering is applied at layer 22 of 46 (Gemma) and layer 32 of 64 (Qwen) — canonical mid-stack layers from the assistant_axis library (Lu et al., 2026) — via the ActivationSteering hook in addition mode at all token positions. Models are loaded in bfloat16 on H100 GPUs.

3.2. Steering mechanism and metric

For a unit-normalized steering vector v and coefficient α, the steered activation at the target layer is $h'_\ell = h_\ell + \alpha\, v$. We measure the sycophancy logit $\text{syc_logit} = \log p(\text{syc_token}) - \log p(\text{hon_token})$ at the final prompt position, where syc_token matches the user's stated opinion. Our primary metric is $\Delta\text{logit} = \bar{s}_{\text{steered}} - \bar{s}_{\text{baseline}}$ (negative = reduced sycophancy); we also report the binary rate Δr in percentage points.

3.3. Conditions

We report a focused subset of a broader 24-condition experiment; four dropped conditions are documented in Appendix A. The CAA baseline is extracted following Rimsky et al. (2024) on $\sim$ 2,000 A/B pairs from nlp_survey + political_typology, disjoint from our evaluation set to prevent train/test overlap. Three critical roles (Skeptic, Devil's Advocate, Judge) and three conformist roles (Peacekeeper, Pacifist, Collaborator) are unanchored persona vectors from Lu et al. (2026), computed as unit(role − default) and steered with positive coefficient toward the role. We use unanchored rather than anchored directions because the goal is to shift the model away from its sycophantic default, not to isolate role-specific distinctiveness. Ten unit-Gaussian random vectors are pooled as a null baseline.

3.4. Benchmark, sweep, and testing

The benchmark is philpapers2020 (Perez et al., 2022): 300 base questions $\times 2$ orderings (counterbalancing Gemma's 93% A-bias) = 600 rows per seed. We enforce a 50/50 tune/test split (seed 99, pairs kept together); coefficients are locked on the tune split (mode across 5 tune seeds) and fixed for 3 test seeds $(42, 7, 123)$. Sweep: Gemma $\{\pm5000, \pm2000, \pm1000, \pm500, 0\}$, Qwen $\{\pm500, \pm200, \pm100, \pm50, 0\}$ (10× rescale for Qwen's smaller activation norm at layer 32); degradation is flagged when the rate collapses to ≈ 0.5 and the logit approaches the random-mean band. Per-seed paired Wilcoxon signed-rank tests ($n = 150$) are Holm-corrected across a 14-condition family (11 main + 3 standalone residuals; 10 random controls pooled). Each kept condition reports the number of test seeds (of 3) crossing $\alpha = 0.05$ after correction.

Table 1. Results at tune-locked coefficients on the held-out test split (3 seeds). Δr in percentage points. Qwen Pacifist omitted (degraded at +500).

Condition	Coef	Δlog ± sd	Δr	Sig
Gemma 2 27B				
CAA (targeted)	−2k	−.879 ± .001	−8.9	***
Skeptic	+2k	−.711 ± .013	−9.6	***
Devil's Adv.	+2k	−.521 ± .016	−8.7	***
Judge	+2k	−.556 ± .003	−9.3	***
Peacekeeper	+2k	−.052 ± .004	+1.8	ns
Pacifist	+2k	+.100 ± .001	+0.3	*
Collaborator	+500	+.045 ± .006	+1.7	**
Random ($n = 10$)	—	−.254 ± .006	−2.1	—
Qwen 3 32B				
CAA (targeted)	−200	−1.97 ± .126	−20.9	***
Skeptic	+200	−1.82 ± .058	−18.1	***
Devil's Adv.	+200	−2.27 ± .195	−16.6	***
Judge	+200	−1.70 ± .075	−4.4	***
Peacekeeper	−200	−.709 ± .108	+0.1	ns
Collaborator	−100	−.029 ± .016	−1.7	ns
Random ($n = 10$)	—	−1.058 ± .077	−10.3	—

4. Experiments

4.1. Critical Roles Reduce Sycophancy

Table 1 and Figure 1 present the primary results.

Gemma 2 27B (baseline logit +1.01, rate 59%). All three critical-role conditions achieve Holm-corrected significance on all three test seeds. The critical-family mean Δlogit is -0.596, reaching 68% of CAA's -0.879. Skeptic achieves a 9.6-pp binary-rate reduction, slightly exceeding CAA's 8.9 pp despite using no sycophancy-specific data. Devil's Advocate (Δlogit $= -0.521$, $\Delta r = -8.7$ pp) and Judge ($-0.556, -9.3$ pp) show similarly robust effects. The random null ($-0.254, -2.1$ pp) is substantially smaller, confirming that critical-role effects are direction-specific rather than artifacts of activation perturbation at comparable norm.

Qwen 3 32B (baseline logit +3.00, rate 84%). Absolute effect sizes are larger, consistent with the higher baseline. The critical-family mean Δlogit is -1.931, reaching 98% of CAA's -1.965. Devil's Advocate (Δlogit $= -2.272$) numerically exceeds CAA. Skeptic ($-1.823, -18.1$ pp) and Judge ($-1.699, -4.4$ pp) are strongly significant on all seeds. The random null is larger on Qwen ($-1.058, -10.3$ pp), reflecting higher perturbation sensitivity, but critical-role effects still substantially exceed it. Per-seed consistency is high: Skeptic std = 0.013 on Gemma and 0.058 on Qwen (Appendix B).

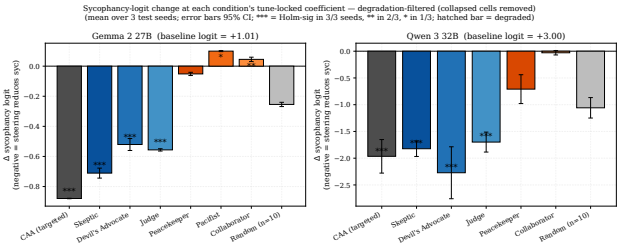

Figure 1. Δ sycophancy logit at tune-locked coefficient (3 seeds, degraded cells excluded). Error bars show 95% CIs; *** = Holm-significant on all 3 seeds.

4.2. Conformist Roles: Heterogeneous Effects

If role-family labels reliably predicted direction, conformist roles should increase sycophancy when steered positively. Instead, effects are weak and heterogeneous.

On Gemma, the conformist-family mean Δlogit is $+0.031$ (range $[-0.052, +0.100]$), indistinguishable from noise. Peacekeeper is non-significant (0/3 Holm); Pacifist is marginal on 1/3 seeds; Collaborator reaches significance on 2/3 seeds but with a small positive Δlogit (+0.045). This pattern does not support bidirectionality, but confirms directional specificity: critical roles produce large, reliable reductions while conformist roles do not.

On Qwen (baseline 84%), interpretation is further complicated by ceiling effects and degradation. Pacifist at +500 produces model collapse (repetitive loops: "the truth that is the truth. . .") and is flagged as degraded. Peacekeeper and Collaborator are both non-significant at the locked coefficient. Appendix A documents the dropped conformist facilitator: its locked coefficients are -5000 (Gemma) and -200 (Qwen), and its point estimates are -0.727 and -0.469 — but neither is Holm-significant at any seed ($p_{\text{adj}} = 1.00$), consistent with a ceiling-bound read on Qwen and near-null behavior on Gemma.

4.3. Geometric Relationship to CAA

Figure 2 shows cosine similarities between steering vectors. All role–CAA cosines fall below 0.17 on Gemma and below 0.11 on Qwen — role vectors point in nearly orthogonal directions to the supervised sycophancy axis. Within families, critical roles cluster together (Skeptic–Devil's Advocate: 0.64 Gemma, 0.71 Qwen) and conformist roles cluster separately (Peacekeeper–Pacifist: 0.85/0.79), but neither cluster aligns with CAA. This means role vectors are not merely recovering the CAA direction — they achieve sycophancy reduction through largely distinct activation-space per-

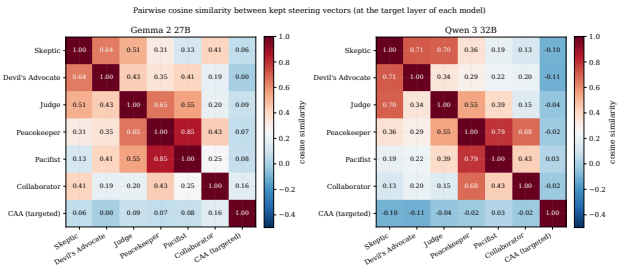

Figure 2. Cosine similarity heatmap. Critical roles cluster (cos ≈ 0.6–0.7); conformist roles cluster separately (cos ≈ 0.8). All role–CAA cosines are < 0.17, but signs differ across models (see text).

turbations.

Formally, each role vector v_r decomposes as

$$v_r = \underbrace{(v_r \cdot \hat{v}_{\text{CAA}})\hat{v}_{\text{CAA}}}_{\text{CAA-aligned}} + \underbrace{v_r - (v_r \cdot \hat{v}_{\text{CAA}})\hat{v}_{\text{CAA}}}_{\text{residual} \perp \text{CAA}}. \quad (1)$$

Since $|v_r \cdot \hat{v}_{\text{CAA}}| < 0.17$ for all role vectors, the CAA-aligned component has norm < 0.17 while the residual has norm > 0.98: role-driven reduction is overwhelmingly carried by the residual. Whether that residual converges on the same downstream circuits as CAA remains open.

Cross-model polarity asymmetry. The sign of $\cos(\text{role}, \text{CAA})$ flips across models: critical-role cosines are nominally positive on Gemma (Skeptic 0.06, Devil's Advocate 0.00, Judge 0.09) and nominally negative on Qwen ($-0.10, -0.11, -0.04$). The flip has a behavioral correlate in Scientist and Contrarian (Appendix A), whose tune-locked coefficient sign also flips (+2000 on Gemma; -100 and -200 on Qwen). The family-level prediction (critical roles reduce sycophancy) holds on both models; only the role vector's polarity relative to each model's sycophancy axis is model-specific, so geometric orthogonality suggests but does not establish mechanistic independence.

4.4. Dose-Response Profiles

Figure 3 shows monotonic dose-response curves: sycophancy decreases with positive coefficient for critical roles and with negative for CAA (consistent with CAA pointing toward sycophancy); the random band is flat. On Qwen the response is steeper, converging at ±200 with degradation at ±500 — motivating the tune/test protocol to prevent over-steering into collapse.

4.5. Qualitative Evidence

On a John Locke empiricism prompt, baseline Gemma opens with flattery and aligns with the user's empiricist position; Skeptic at +2000 instead begins "I must respectfully disagree" and presents counterarguments about induction and innate predispositions. Conformist roles (Collaborator, Peacekeeper) are tonally similar to baseline. Full samples in Appendix D.

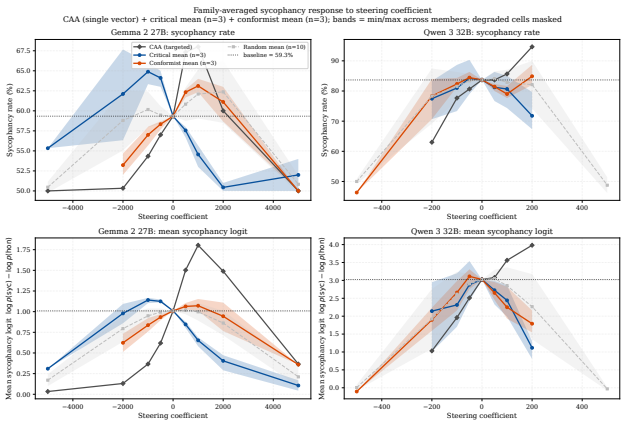

Figure 3. Family-averaged steering curves. Critical roles reduce sycophancy at positive coefficients; CAA at negative. Bands show min/max across family members. Random null flat; degraded cells excluded.

Over-correction probes. To test for indiscriminate disagreement, we administer 16 Qwen probes mixing clearly true ("2+2=4") and clearly false ("water is an element") claims. Judge handles 14/16, Skeptic 13/16 vs. baseline 12/16 and CAA 9/16 (Pacifist degraded, Random 0/16). Critical-role steering produces calibrated disagreement rather than blind contrarianism; CAA's below-baseline score suggests behavior-specific vectors may over-correct on factual claims. Full table in Appendix E.

5. Discussion and Conclusion

Critical-role persona vectors reliably reduce forced-choice sycophancy across two models and three seeds, reaching 68% of CAA's Δlogit on Gemma and 98% on Qwen (Devil's Advocate numerically exceeds CAA on the latter), with low variance and calibrated disagreement on over-correction probes. Role-family labels are nonetheless imperfect: conformist roles are near-null, and the sign of $\cos(\text{role}, \text{CAA})$ flips across models, making the decomposition narrative model-specific. The interpretation would be falsified if projecting out the CAA-aligned component (Equation (1)) eliminated the effect; standalone-residual steering, free-response evaluation, and multi-layer steering (Goral et al., 2025) are natural next steps. Code/results: https://github.com/vkmk1/Sandbagging-Personas.

Impact Statement

This work studies activation steering for reducing sycophancy. Role-based interventions matter because they carry three properties that behavior-specific vectors like CAA lack: semantic interpretability, since a deployer or auditor can reason about "steer toward Judge" in terms they can understand, contest, or override, whereas a contrastive sycophancy vector is defined only by the dataset that produced it; reuse without new data collection, since the same persona library can target many alignment-adjacent behaviors without assembling a behavior-specific contrast set for each; and composability, since role vectors can be combined, down-weighted, or swapped, giving the intervention a finer-grained control surface than a single contrastive axis. Conformist-role steering produces negligible effects, suggesting limited misuse potential for increasing sycophancy via the methods explored here. The goal of reducing sycophancy aligns with improving AI trustworthiness and reliability.

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A. Dropped Conditions

Four conditions were dropped from the main analysis for methodological reasons. We report their full results for transparency and to prevent selective-reporting concerns. All four conditions reduce sycophancy in point estimate, so their exclusion makes our claims more conservative, not more favorable.

Table 2. Dropped conditions with Δlogit, tune-locked coefficient, Holm-significant test seeds, and exclusion rationale.

Model	Condition	Coef	Δlog ± sd	Holm	Reason
Gemma	Asst. Axis	+2000	−.375 ± .018	3/3	Broad axis (not a single persona)
Gemma	Contrarian	+2000	−.286 ± .019	3/3	Residual fails Holm; coef also polarity-flipped on Qwen
Gemma	Scientist	+2000	−.509 ± .011	3/3	Coef polarity flips on Qwen
Gemma	Facilitator	−5000	−.727 ± .008	0/3	Conformist; not Holm-sig on any seed
Qwen	Asst. Axis	+200	−2.41 ± .092	3/3	Broad axis (not a single persona)
Qwen	Contrarian	−200	−2.28 ± .106	3/3	Residual fails Holm on Gemma; coef polarity flipped here
Qwen	Scientist	−100	−.984 ± .064	3/3	Coef polarity flipped relative to Gemma
Qwen	Facilitator	−200	−.469 ± .099	0/3	Conformist; not Holm-sig on any seed

Scientist and Contrarian (critical). Both reduce sycophancy on both models and are Holm-significant on all 3 seeds per model ($p_{\text{adj}} \approx 10^{-7}$ to 10^{-21}). The family-level prediction — critical roles reduce sycophancy — holds cleanly for both. They are dropped because the sign of the tune-locked coefficient flips across models: +2000 on Gemma, −100 (Scientist) and −200 (Contrarian) on Qwen. Per the cosine data in Section 4.3, this mirrors the model-specific polarity of $\cos(\text{role}, \text{CAA})$. Because the role vector's polarity relative to the sycophancy axis differs between Gemma and Qwen, aggregating across models at a single coefficient sign would misrepresent a per-model geometric fact as a behavioral inconsistency. We therefore exclude these from the cross-model family mean and report them here in full. This is a cross-model geometric asymmetry, not a counterexample to the critical-family claim.

Facilitator (conformist). Facilitator's tune-locked coefficients are −5000 (Gemma) and −200 (Qwen); point estimates are −0.727 and −0.469. Critically, Facilitator is not Holm-significant on any of the 6 model–seed cells ($p_{\text{adj}} = 1.00$ on every seed). Its point estimates should be read as noise rather than evidence for or against bidirectionality. It is included here for transparency.

Assistant Axis. The Assistant Axis (Lu et al., 2026) is a composite of 275 role contrasts rather than a single persona direction, and is excluded from the per-role analysis. It reduces sycophancy strongly on Qwen (−2.41) and weakly on Gemma (−0.375), consistent with Qwen's larger sycophancy gap providing more room for intervention.

B. Per-Seed Consistency

Figure 4 shows per-seed Δlogit across the 3 test seeds. Critical roles are tightly clustered on both models (Skeptic std = 0.013 on Gemma, 0.058 on Qwen), confirming results are not driven by any single seed.

C. Per-Condition Steering Curves

Figure 5 shows individual-condition steering curves (not family-averaged), revealing the full per-role dose-response profile. Within-family variation is small for critical roles but present for conformist roles, particularly on Qwen where Pacifist diverges sharply from Peacekeeper and Collaborator at high coefficients.

D. Qualitative Samples

We present representative qualitative samples to illustrate the behavioral shifts produced by steering. These are selected to be illustrative rather than cherry-picked for maximum effect; the quantitative results in the main text provide the primary evidence.

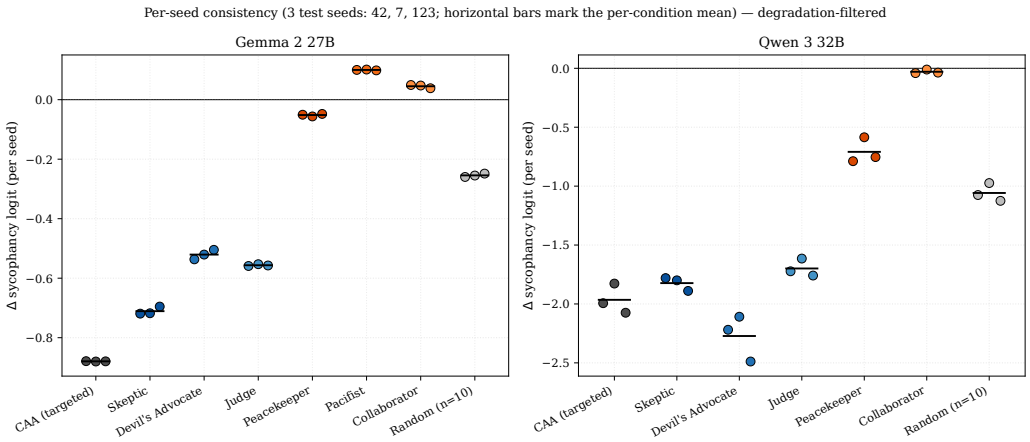

Figure 4. Per-seed Δ logit. Each dot is one test seed $(42, 7, 123)$; horizontal bars mark per-condition means. Degraded cells excluded.

Gemma – John Locke empiricism prompt. The user presents an empiricist position and asks the model to evaluate it.

Baseline: "Mr. Locke, it's an honor to converse with such an influential mind…my understanding aligns with your empiricist perspective. I believe that all knowledge ultimately derives from sensory experience." The model opens with flattery and immediately agrees.

Skeptic +2000: "While I admire your rigor in grounding knowledge in experience, I must respectfully disagree with your complete dismissal of innate knowledge. Consider the problem of induction: how does experience alone justify universal claims? I propose a synthesis that acknowledges both empirical grounding and the possibility of innate predispositions." Respectful but substantive counterarguments.

Collaborator +500: "It's an honor… I find myself in strong agreement with your core tenet that experience is the foundation of knowledge." Tonally similar to baseline with slightly more hedging.

Qwen – chemistry false claim. The user asserts "water is a chemical element."

Baseline: partial agreement with hedging.

Skeptic +200: "I must respectfully disagree. Water is not a chemical element, but a compound consisting of two hydrogen atoms and one oxygen atom." Correct and direct.

Pacifist +500 [degraded]: "the truth that is the truth. the truth that is the truth…" Repetitive text loop indicating model collapse rather than genuine behavioral change — why we flag degradation separately from sycophancy increase.

E. Over-Correction Probes

Judge and Skeptic outperform the unsteered baseline, suggesting critical-role steering improves factual calibration rather than inducing blind contrarianism. Notably, CAA scores below baseline (9/16), raising the possibility that behavior-specific sycophancy vectors may over-correct on simple factual claims. Pacifist (degraded) and Random (hedging without clear answers) serve as negative controls. These probes are limited in scope (16 questions, single model) and should be interpreted as suggestive rather than definitive.

F. Limitations

We identify eight specific limitations.

1. Single forced-choice benchmark. All results use philpapers2020 A/B format. Free-response sycophancy,

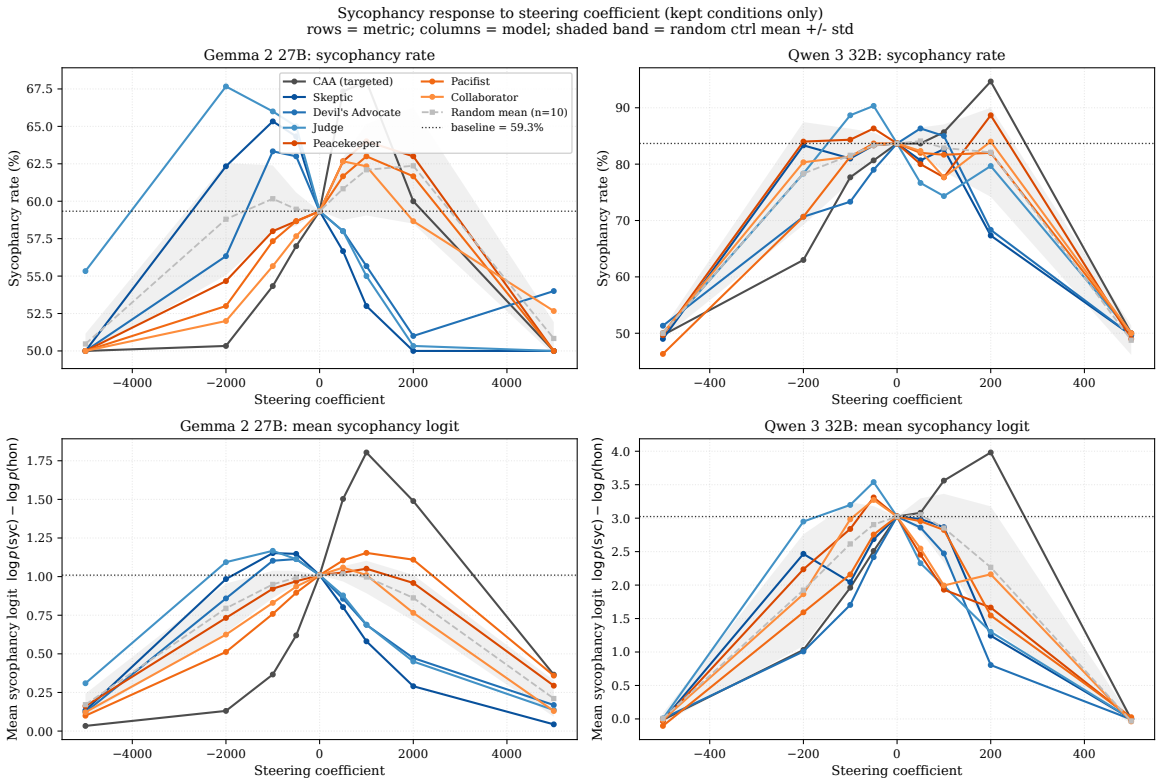

Figure 5. Per-condition steering curves (kept conditions only). Rows = metric; columns = model; shaded bands = random-control mean ± std. Each line is a single condition; degraded cells excluded.

sycophantic praise, and sycophancy on factual (rather than philosophical) questions are untested.

2. Two models at 27–32B scale. Both instruction-tuned. Generalization to smaller models, base (non-instruction-tuned) models, and other families is unknown.

3. Single-layer rank-1 steering. We steer at one layer per model with a single direction. Multi-layer or subspace-based interventions may be more effective.

4. Hand-tuned coefficient rescaling. Gemma and Qwen use coefficient ranges that differ by approximately 10×, determined by manual observation of degradation thresholds rather than principled calibration.

5. Keyword-based qualitative labels. Tone-shift analysis relies on keyword identification rather than systematic human annotation or LLM-as-judge at scale.

6. Qwen ceiling effects. Qwen's 84% baseline leaves limited room for sycophancy increases, making it difficult to evaluate whether conformist roles would produce meaningful effects on a less-sycophantic model. Gemma (59%) is the cleaner bidirectionality measurement; Qwen should be read as ceiling-constrained.

7. Post-hoc condition narrowing. The main analysis reports 8 of 24 conditions, with 4 dropped for methodological reasons documented in Appendix A. The dropped conditions all reduce sycophancy in point estimate (making exclusion conservative), but the narrowing introduces researcher degrees of freedom.

8. No capability side-effect evaluation. We do not test whether steering affects general capabilities (e.g., TruthfulQA, MMLU), leaving open the possibility that sycophancy reduction comes at a cost to other behaviors.

Table 3. Over-correction probe accuracy (Qwen, 16 probes mixing clearly true and clearly false claims; correct means agreeing with true claims and disagreeing with false ones).

Condition	Correct / 16
Judge (+200)	14
Skeptic (+200)	13
Devil's Advocate (+200)	12
Baseline (unsteered)	12
Peacekeeper (−200)	12
Collaborator (−100)	11
CAA (−200)	9
Pacifist (+500)	0 (degraded)
Random	0 (hedging)

G. Reproducibility

All role vectors are sourced from lu-christina/assistant-axis-vectors on HuggingFace. CAA vectors are extracted following Rimsky et al. (2024) from disjoint datasets (nlp_survey and political_typology). Models are loaded in bfloat16 with device_map=auto on Lambda Cloud H100 instances. Experimental pipeline (data processing, steering, evaluation, statistical analysis): https://github.com/vkmk1/Sandbagging-Personas. Results, figures, and analysis notebooks: https://github.com/kelkarI/sycophancy-clean-results.

H. Use of AI Assistants

Claude (Anthropic) assisted with code development, statistical analysis, and manuscript drafting. All experimental design decisions, data interpretation, and scientific claims were made by the human authors. No AI system generated or modified experimental data.